29/10/2025

D.T.F.T.

Synthesis $\longrightarrow x[n] = \frac{1}{2\pi}\int_{2\pi} X(e^{j\omega}) e^{j\omega n} d\omega$

Analysis eqn. $\longrightarrow X(e^{j\omega}) = \sum_{n=-\infty}^{\infty} x[n] e^{-j\omega n}$

Time domain

Frequency domain

Alternate representation of the signal

$x[n] \rightleftharpoons X(e^{j\omega})$

C.T.F.T.

$X(j\omega)$

ω takes unique values from $-\infty$ to ∞

$x(e^{j\omega})$

$\Rightarrow x(e^{j0})$

$= x(e^{j2\pi})$

$= x(e^{jk\cdot 2\pi})$

Multiplication in time $\rightleftharpoons$ Conv in frequency

$x[n]\,h[n] \rightleftharpoons \frac{1}{2\pi} X(e^{j\omega}) * H(e^{j\omega})$

$y[n] = x[n] * h[n] \rightleftharpoons X(e^{j\omega})H(e^{j\omega})$

Ex Say our system is an LTI system and is a delay unit.

$h[n] = \delta[n-n_0]$

$y[n) = x[n] * h[n] = x[n] * \delta[n-n_0] = x[n-n_0]$

$x[n] \rightleftharpoons X(e^{j\omega})$

$h[n] \rightleftharpoons e^{-j\omega n_0}$

$Y(e^{j\omega}) = e^{-j\omega n_0} X(e^{j\omega})$

$y[n] = x[n-n_0]$

Ex i/p $\Rightarrow x[n] = \alpha^n u[n]$ $\quad 0<\alpha<1$

Impulse response $\Rightarrow$ of an LTI system $h[n] = \beta^n u[n]$ $\quad 0<\beta<1$

Compute $y[n]$.

$x[n] = \alpha^n u[n]$

$$X(e^{j\omega}) = \sum_{n=-\infty}^{\infty} \left(\alpha^n u[n]\right) e^{-j\omega n}$$

$$= \sum_{n=0}^{\infty} \left(\alpha e^{-j\omega}\right)^n$$

$$= \frac{1}{1-\alpha e^{-j\omega}}$$

Similarly,

$$H(e^{j\omega}) = \frac{1}{1-\beta e^{-j\omega}}$$

$$Y(e^{j\omega}) = \frac{1}{(1-\alpha e^{-j\omega})(1-\beta e^{-j\omega})}$$

$$= \frac{A}{1-\alpha e^{-j\omega}} + \frac{B}{1-\beta e^{-j\omega}}$$

$$= A\alpha^n u[n] + B\beta^n u[n]$$

Write A & B explicitly in terms of α & β.

Let $\alpha = \beta$.

$$Y(e^{j\omega}) = \frac{1}{(1-\alpha e^{-j\omega})^2}$$

$$\frac{d}{d\omega}\left(\frac{1}{1-\alpha e^{-j\omega}}\right) = \frac{-\left(-\alpha \cdot e^{-j\omega} \cdot -j\right)}{(1-\alpha e^{-j\omega})^2}$$

$$j\frac{d}{d\omega}\left(\frac{1}{1-\alpha e^{-j\omega}}\right) = \frac{1}{(1-\alpha e^{-j\omega})^2}$$

$e^{-j\omega} \cdot \alpha$

$$\frac{e^{j\omega}}{\alpha} \cdot j\left[\frac{d}{d\omega}\left(\frac{1}{1-\alpha e^{-j\omega}}\right)\right] = \frac{1}{(1-\alpha e^{-j\omega})^2}$$

$$\alpha^n u[n] \rightleftharpoons \frac{1}{1-\alpha e^{-j\omega}}$$

$$n\alpha^n u[n] \rightleftharpoons j\frac{d}{d\omega}\left[\frac{1}{1-\alpha e^{-j\omega}}\right]$$

$$\frac{n\alpha^n u[n]}{\alpha} \rightleftharpoons \frac{j}{\alpha}\frac{d}{d\omega}\left[\frac{1}{1-\alpha e^{-j\omega}}\right]$$

$$\frac{(n+1)\alpha^{n+1} u[n+1]}{\alpha} \rightleftharpoons \frac{e^{j\omega}}{\alpha} j\frac{d}{d\omega}\left[\frac{1}{1-\alpha e^{-j\omega}}\right]$$

$$(n+1)\alpha^n u[n+1] \rightleftharpoons \left(\frac{1}{1-\alpha e^{-j\omega}}\right)^2$$

$$X(e^{j\omega}) = \sum_{n=-\infty}^{\infty} x[n] e^{-j\omega n}$$

$$\frac{d}{d\omega}X(e^{j\omega}) = \sum_{n=-\infty}^{\infty} x[n](-jn) e^{-j\omega n}$$

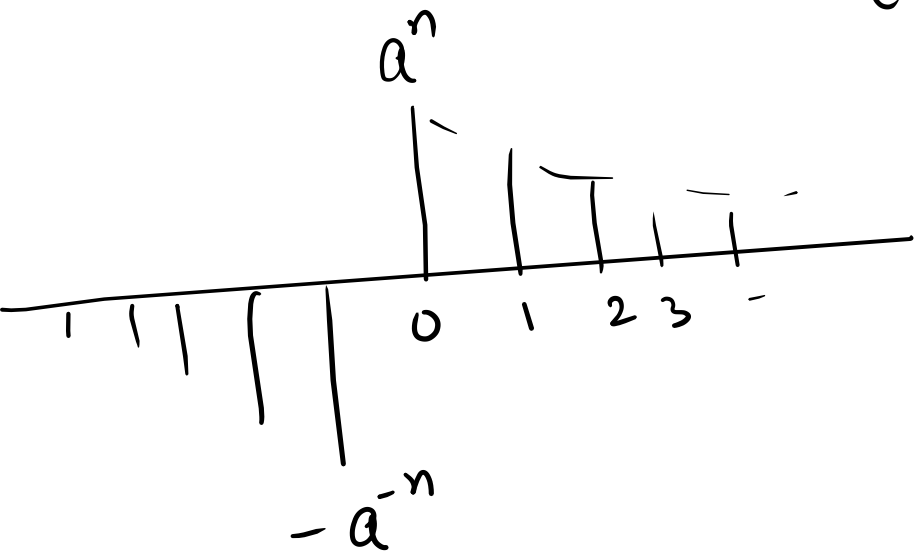
$$j\frac{d}{d\omega}X(e^{j\omega}) = \sum_{n=-\infty}^{\infty} (nx[n]) e^{-j\omega n}$$

$$x[n] \rightleftharpoons X(e^{j\omega})$$

$$x[n+1] \rightleftharpoons e^{j\omega} X(e^{j\omega})$$

$e^{j\omega n_0}$

$$u[n] \rightleftharpoons \frac{1}{1-e^{-j\omega}} + \pi\delta(\omega)$$



S: $y[n] = \sum_{k=-\infty}^{n} x[k]$

$= x[n] * h[n]$

where $h[n] = u[n]$

$$y[n] = \sum_{k=-\infty}^{\infty} x[k]\, h[n-k]$$

$$= \sum_{k=-\infty}^{\infty} x[k]\, u[n-k]$$

$$= \sum_{k=-\infty}^{n} x[k]$$

$u[n-k] = 1$ for $n-k \geq 0$

$n \geq k$

Or $k \leq n$

$$Y(e^{j\omega}) = X(e^{j\omega})\,H(e^{j\omega})$$

$$= X(e^{j\omega})\;U(e^{j\omega}) \leftarrow \text{For an accumulator}$$

$$= X(e^{j\omega})\left[\frac{1}{1-e^{-j\omega}} + \pi\delta(\omega)\right]$$

$$= \frac{X(e^{j\omega})}{1-e^{-j\omega}} + \pi X(e^{j0})\delta(\omega)$$

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in D.T. $\quad x[n] \rightleftharpoons X(e^{j\omega})$

Accumulator is a counter-part of integrator in C.T. $\Longrightarrow \sum_{k=-\infty}^{n} x[k] \rightleftharpoons \frac{X(e^{j\omega})}{1-e^{-j\omega}} + \pi X(e^{j0})\delta(\omega)$

$$x[n] \rightleftharpoons X(e^{j\omega})$$

$$x[n] - x[n-1] \rightleftharpoons (1 - e^{-j\omega}) X(e^{j\omega})$$

S: $y[n] = \alpha_1 x[n] + \alpha_2 x[n-1]$

$$Y(e^{j\omega}) = (\alpha_1 + \alpha_2 e^{-j\omega}) X(e^{j\omega})$$

$$H(e^{j\omega}) = \frac{Y(e^{j\omega})}{X(e^{j\omega})} = \alpha_1 + \alpha_2 e^{-j\omega}$$

$$h[n] = \alpha_1 \delta[n] + \alpha_2 \delta[n-1]$$